DATA 2010 Project Report: Trends in Olympic Athlete Performance and Representation (1896–2024)
Winter 2026

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1 Project Overview

Research Question: The Olympic Games provide a long-running record of athlete participation and medal outcomes, allowing us to study how performance and representation evolve over time. Using the *Olympic Summer and Winter Games (1896–2024)* dataset (athlete–event records with variables such as `year`, `season`, `sports`, `event`, `team`, `sex`, `age`, and `medal`), this project investigates three connected questions. First, we examine how medal outcomes vary by **age group** and **sex**, and how **gender parity** and **typical medal-winning age** changed across decades. Second, we evaluate how country participation size is associated with medal counts, medal efficiency, and Olympic dominance. Third, we examine which athlete-level factors most influence the probability of winning a medal and whether country/team membership adds predictive value after accounting for athlete characteristics.

Why it matters: The answers matter to stakeholders who make decisions about athlete development and sport investment. National sport organizations and coaches can use age- and sex-specific medal patterns to support talent pipeline planning. Analysts, journalists, and policymakers can benefit from fairer country comparisons by understanding how medal totals depend on delegation scale and efficiency rather than raw medal counts alone. Finally, understanding whether country-level advantages remain important after accounting for athlete characteristics can help explain why some countries outperform others over time.

Main Results: Overall, the project shows that Olympic outcomes are shaped by both athlete-level and country-level factors. First, female participation increases substantially across decades and moves much closer to parity, while medalist composition and typical medal-winning ages vary across age groups and sex. Second, country-year participation size is strongly positively associated with event-medal counts, but simple regression is not sufficient on its own; log transformation and multiple regression show that broader sport participation and time trends also help explain medal outcomes. Third, medal success is naturally a binary classification problem, and logistic regression shows that age, sex, sport, and country/team membership all contribute to an athlete-event row's probability of winning a medal.

The Olympic Games are not only a global competition but also a long-running dataset that can reveal how representation, performance, and strategic investment in sport evolve over time. Using Olympic records spanning 1896–2024, this project investigates patterns with real-world relevance to athlete development and national sport planning. Specifically, we focus on three research questions:

The Three Main Research Questions:

1) ***How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?***

2) ***To what extent is country participation size associated with event-medal counts, and how does this relationship shape medal efficiency and Olympic dominance?***

3) ***What factors most influence an athlete's probability of winning a medal, and to what extent does country/team membership contribute to medal probability after accounting for athlete characteristics?***

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2 Data Source and Description

2.1 Where the data came from:

- **Dataset name:** Olympic Summer and Winter Games (1896–2024)

- **Link (Kaggle):** https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024

- **Dataset owner (as shown on Kaggle):** Cassini-Chris (Owner)

- **Authors:** Not specified on the Kaggle dataset page

- **Provenance:** Not specified on the Kaggle dataset page

- **Citations / DOI:** Not specified on the Kaggle dataset page

- **What each row represents:** One row represents one athlete's record in one Olympic event for a specific Olympic Games, including the athlete's sex/age, country/team, the sport and event, the Games year/season/city, and the medal outcome (if any).

- **Key columns used (2–8 columns):**
  - `year`
  - `season`
  - `team`
  - `sports`
  - `event`
  - `sex`
  - `age`
  - `medal`

- **License:** Apache License 2.0

- **Expected update frequency:** Not specified on the Kaggle dataset page (Updated 2 years ago)

2.2 Basic dataset facts

Number of rows	290,697
Number of columns	8
Time range (Year)	1896–2024
Seasons included	Summer & Winter
Unique years	38
Unique teams (`team`)	1151
Unique sports (`sports`)	92
Unique events (`event`)	1534
Medalist (Gold/Silver/Bronze)	42756
Non-medalist	247,941
Age range (years)	10–72
Sex proportion (M/F)	69.38% / 30.62%
Missing values present?	No, there is no missing values in the entire dataset
Exact duplicate rows removed	0

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3 Data Cleaning and Feature Engineering

This section documents how the Olympics dataset was prepared for analysis and how new variables were created to support the three research questions.

Minimum Expectations:

- Missing-value handling was included in the data-cleaning process, although no missing values were present in the dataset.

- Removed duplicates and impossible values. As part of the data-cleaning process, we checked for duplicate and impossible values, but none were found in the dataset.

- Dealt with outliers *carefully* (flagged suspicious ages instead of deleting them).

What actually applied:

- Standardized column names and stripped whitespace from text fields such as `team`, `sports`, `event`, and `medal`.

- Standardized medal labels and created a binary variable `won_medal` indicating whether an athlete-event row resulted in Gold, Silver, or Bronze.

- Engineered ordered `age_group` categories so age-based comparisons would be easier to interpret in plots and tables.

- Created `medal_points` for descriptive summaries comparing overall medal success.
- Cleaned detailed sport labels and grouped them into broader `sport_category` values to reduce inconsistency and improve interpretability in RQ3.
- Cleaned country/team labels and created `team_clean` so country-level aggregation would be consistent.
- Created an *event-medal* view by deduplicating medals at (`year`, `event`, `team_clean`, `medal`) so that team events would not inflate country-level medal totals.
- Built country-year summaries including participation size, event-medal counts, efficiency, number of sports, and year for use in RQ2 regressions.

Show evidence (before vs. after):

Check	Before cleaning	After cleaning
Country/team labels	historical names, spelling variation, and non-country teams	cleaned to `team_clean`
Sport labels	overly detailed / inconsistent	standardized and grouped into `sport_category`
Medal labels	raw medal values / no binary medal indicator	standardized and used to create `won_medal`
Age representation	raw age values only	ordered `age_group` created for analysis
Country medal counting	team events could inflate country medal totals	event-medal view created to deduplicate country-level medals
Country-year analysis data	no country-year summary table	participation size, event medals, efficiency, number of sports, and year grouped and summarized

4 Exploratory Data Analysis and Visualization

Research Question 1:

How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?

Each research question is divided into sub-questions, and each sub-question maps to a specific plot(s), ensuring that every visualization directly supports the main research question.

4.1 How do medal outcomes vary by age group and sex?

Medalist Share by Age Group (100% stacked by sex)

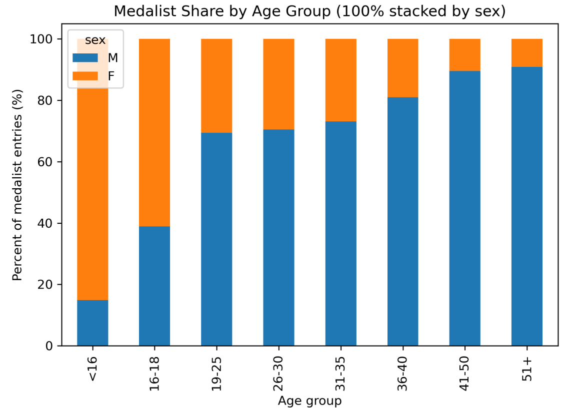

Figure 1: Each bar sums to 100%, showing the female vs. male share of medalist entries within each age group. This makes comparisons fair even when some age groups have far fewer medalist entries.

This figure indicates that sex balance among medalists varies by age group. Some age groups are more male-dominated, while others are closer to balanced. Differences are influenced by sport mix and historical changes in female participation.

4.2 Which decades show the largest changes in gender parity?

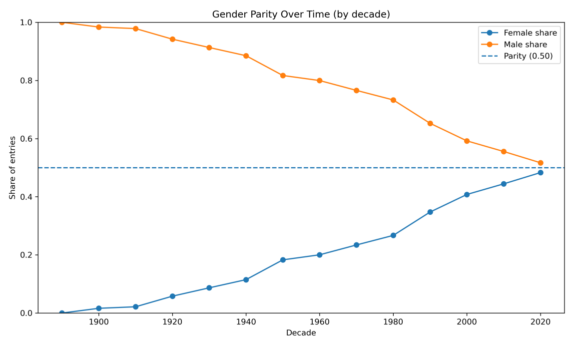

Figure 2: Female and male participation shares by decade. The dotted 0.50 line represents parity.

This plot shows a clear long-term increase in female participation over decades, moving toward parity. It provides strong evidence that Olympic representation has become more gender-balanced over time.

Limitations: This is participation share, not medal share. Dataset composition and coverage across early editions can affect decade-level estimates.

4.3 Largest Decade-to-Decade Changes in Female Share

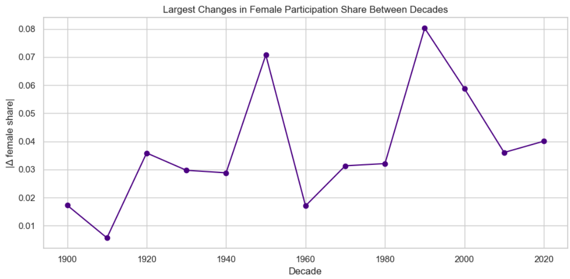

Figure 3: Absolute changes in female participation share between consecutive decades. Peaks identify decades with the largest shifts in gender representation.

This plot identifies the decades where gender representation changed the most. The highest peaks highlight key transition periods likely associated with expanded women's events and broader inclusion policies.

4.4 How do peak-age patterns change across decades (median and IQR of medalist ages)?

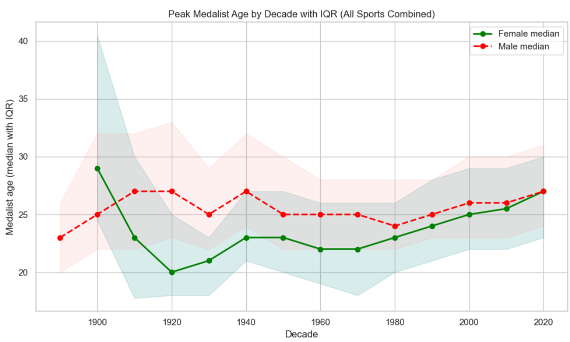

Figure 4: Median medalist age by decade with IQR shading (25th–75th percentile), computed across all sports combined and split by sex.

This figure summarizes how the typical medalist age and its spread change across decades for females and males. If medians rise, typical medal-winning ages trend older; if IQR widens, medalist ages become more varied.

4.5 RQ1 Conclusion, Limitations and Next Steps

Conclusion: Overall, the results show that the Olympics become substantially more gender-balanced over time, with female participation increasing steadily and moving closer to parity in recent decades. Medalist composition varies across age groups and sex, and the peak-age plot indicates that typical medalist ages and their variability shift across decades.

Limitations: The plot uses entries of medalists instead of all participants, meaning team events can inflate counts. Participation share is shown instead of medal share due to the composition of the dataset.(Can mask specific peak ages). Coverage of the earlier Olympic editions can affect the estimation at the decade level as well as instances of missing events such as the 2022 Summer Olympics as well during waring times like 1916, 1940 and 1944.

Next steps: We would repeat the peak-age analysis by individual sports, separate Summer vs. Winter editions, include decade sample sizes on plots, and extend beyond EDA to a simple predictive model for medal probability.

Research Question 2: To what extent is country participation size associated with event-medal counts, and how does this relationship shape medal efficiency and Olympic dominance?

4.6 Is there a positive relationship between the number of athletes a country sends and the number of medals it wins?

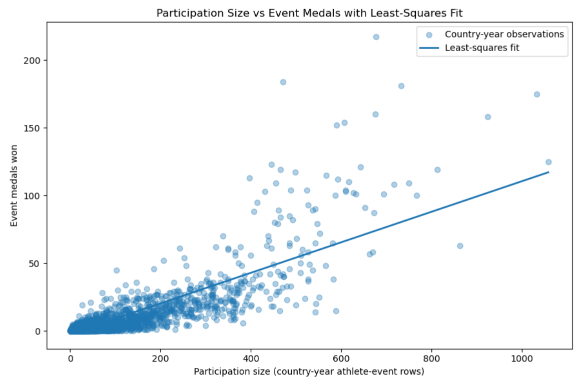

Figure 5: Scatterplot of country-year participation size versus event-medal counts, with a least-squares regression line summarizing the overall relationship.

This plot shows a strong positive association between participation size and event medals won. In general, country-years with larger delegations tend to win more medals.

The least-squares line captures this overall upward trend, but the points are fairly spread out around it. That spread becomes larger for bigger delegations, showing that countries with similar participation sizes can still have quite different medal totals. A few very large, high-medal observations also stand out, so the relationship is positive but not perfectly linear.

Limitations:

- Participation size is measured using athlete-event rows, not unique athletes.
- The variability increases at higher participation levels, so the fit is less consistent for large delegations.
- A small number of extreme country-year observations may pull the regression line upward.

Log-Transformed Participation Size vs Log-Transformed Event Medals

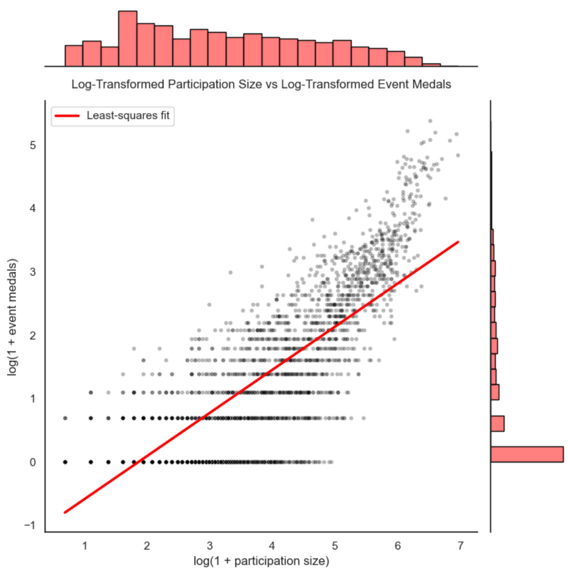

Figure 6: Log-transformed scatterplot showing the relationship between participation size and event-medal counts after reducing skewness and improving visibility at low values.

This log-transformed plot shows the relationship more clearly than the raw-scale scatterplot. After applying log(1 + participation size) and log(1 + event medals), the dense cluster of small country-year observations is spread out, making the overall pattern easier to see.

The points follow a clearer positive linear trend, and the least-squares line passes more centrally through the cloud. This suggests that the log transformation gives a better linear summary of the association between participation size and medal output. It also reduces the visual influence of extremely large delegations, so the relationship appears more stable across the full range of observations.

Overall, this figure supports the idea that participation size and medal success are positively associated, and that the relationship is represented more effectively on the log scale.

Limitations:

- There is still noticeable scatter around the fitted line, so participation size alone does not fully explain medal outcomes.
- The horizontal banding remains because medal counts are discrete values.
- The transformation improves linearity and visibility, but interpretation is now on the log scale, not the original scale.

Residual Plot for the Multiple Regression Model

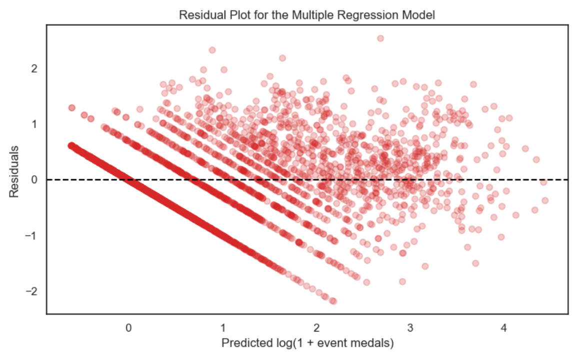

Figure 7: Residual plot from the multiple regression model, showing how prediction errors vary across fitted values.

This residual plot suggests that the multiple regression model improves on the simpler model, but it still does not fit perfectly. Many residuals are centered near zero, which is a good sign, and the spread is more balanced than in the one-predictor model.

At the same time, the residuals do not form a fully random cloud. There is clear diagonal banding and some remaining structure, especially across the middle range of predicted values, which shows that the model still leaves systematic variation unexplained. This supports the idea that adding predictors like participation size, number of sports, and year improves the fit, but does not completely capture medal outcomes.

Overall, the plot suggests that the multiple regression model is a better summary of the relationship for RQ2, though it is still only an approximation.

Limitations:

- The diagonal banding remains because the response is based on discrete medal counts, even after log transformation.
- Residual spread is still noticeable at moderate and higher fitted values, so the error variance is not fully constant.
- Important factors such as country strength, host advantage, and sport composition are not fully captured by the model.

4.7 Which countries are the most efficient at converting participation into medals?

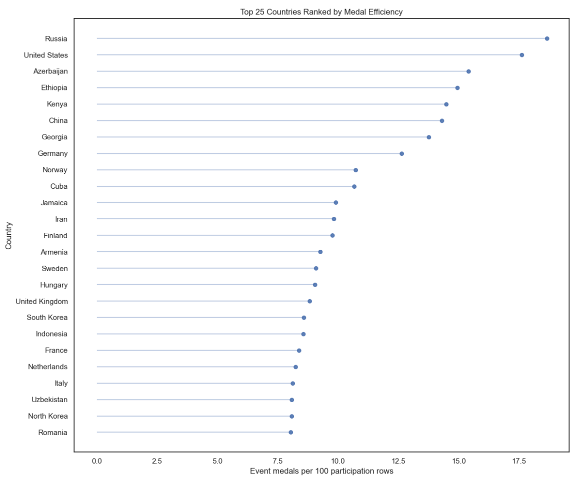

Figure 8: Countries ranked by medal efficiency, measured as event medals per 100 participation rows, after filtering out very small delegations.

This ranking compares countries by how efficiently they turn participation into medals after applying the minimum-size filter. Countries at the top win the most event medals per 100 participation rows, meaning they achieve stronger medal returns relative to the size of their Olympic participation.

The ranking shows that efficiency is not simply the same as delegation size. Some countries perform very well because they convert a moderate level of participation into a high medal yield, which may reflect strong specialization or consistently strong performances in the sports they enter. At the same time, highly efficient countries are not always the ones with the largest overall medal totals, so this plot highlights relative return rather than absolute dominance.

Overall, the figure suggests that some countries are especially effective at converting participation opportunities into medals.

Limitations:

- Efficiency is measured using athlete-event rows, not unique athletes.
- The ranking depends on the minimum participation threshold, so countries included or excluded can change the results.
- A high efficiency score does not necessarily mean broad success across many sports; it may reflect concentration in a smaller set of strong events.

4.8 Which countries dominate the Olympic Games in a given year, and is this dominance primarily driven by delegation size or efficiency?

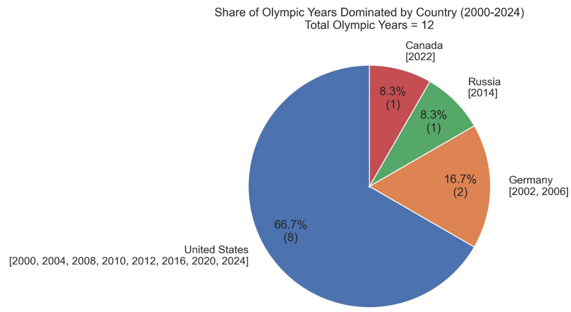

Figure 9: Each section shows the percentage share of Olympic years from 2000 to 2024 in which a given country had the highest event-medal count.

This plot summarizes which countries most often dominated the Olympic Games from 2000 to 2024, based on having the highest event-medal count in a given Olympic year. The pattern is strongly concentrated rather than broadly shared: the United States accounts for most of these top finishes, while Germany, Russia, and Canada represent much smaller shares.

Because the figure aggregates dominance across years, it emphasizes how unevenly Olympic leadership is distributed over the period. The large share for the United States shows that recent Olympic dominance was not spread widely across many countries. Instead, it was concentrated in a small group of nations, with one country leading far more often than the others.

Overall, the figure suggests that Olympic dominance from 2000 to 2024 was driven by a limited set of countries rather than rotating broadly across the field.

Limitations:

- The dataset is missing the 2018 Winter Olympics, so the timeline is incomplete.
- The plot shows only the top country in each year, not how far ahead it was of the next countries.
- Summer and Winter Olympics are combined, so top medal counts are not directly comparable across all years.

4.9 RQ2 Conclusion and Next Steps

Conclusion (RQ2): Overall, the results show that Olympic medal success is strongly associated with participation size, with countries that enter more athlete-event rows generally winning more event medals. The log-transformed analysis makes this relationship clearer and more stable, while the multiple regression suggests that participation alone is not enough to explain medal outcomes fully. The efficiency rankings show that some countries convert participation into medals especially well, but the year-by-year dominance plot suggests that repeated leadership is still concentrated among a small group of countries. Taken together, the results suggest that both scale of participation and efficiency of conversion matter, but large and historically strong programs remain the most likely to dominate medal tables over time.

Next steps (RQ2): We would separate Summer and Winter Olympics, compare top-country dominance using relative gaps rather than just first-place counts, and test alternative models better suited for count outcomes. We would also add predictors such as host-country status, sport mix, and country fixed effects to better explain why countries with similar participation sizes can have very different medal returns.

Research Question 3: What factors most influence an athlete's probability of winning a medal, and to what extent does country/team membership contribute to medal probability after accounting for athlete characteristics?

4.10 How do age, sex, and sport affect an athlete's probability of winning a medal?

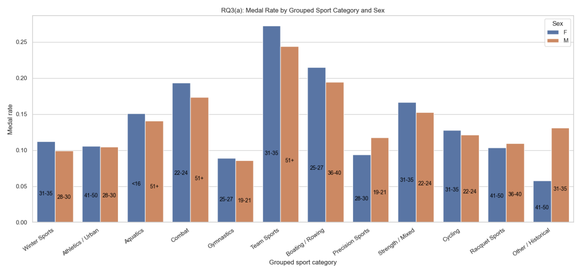

Figure 10: Medal rate by grouped sport category and sex, with dominant medal-winning age-bin labels added to each bar.

This plot shows that medal rates differ clearly across grouped sport categories and by sex. Team Sports, Boating/Rowing, and Combat have some of the highest medal rates, while categories such as Other/Historical, Precision Sports, and Gymnastics are lower. In most categories, the female and male bars are fairly close, but some gaps remain, showing that the relationship between sex and medal rate is not identical across all sports.

The age labels also suggest that the most common medal-winning age range varies by sport category. For example, some sports peak in younger adult ranges, while others are more concentrated in older age bins. This supports the idea that medal probability depends not just on sex, but also on the type of sport and the age profile typical for success in that sport.

Limitations:

- The analysis is based on athlete-event rows, not unique athletes.
- Grouping sports improves clarity, but it hides differences within each category.
- The plot is descriptive; the size of a bar difference alone does not show whether the difference is statistically meaningful.

4.11 After controlling for age, sex, and sport, which countries/teams still exhibit higher or lower baseline medal probability, and does country/team membership explain meaningful variation in medal outcomes?

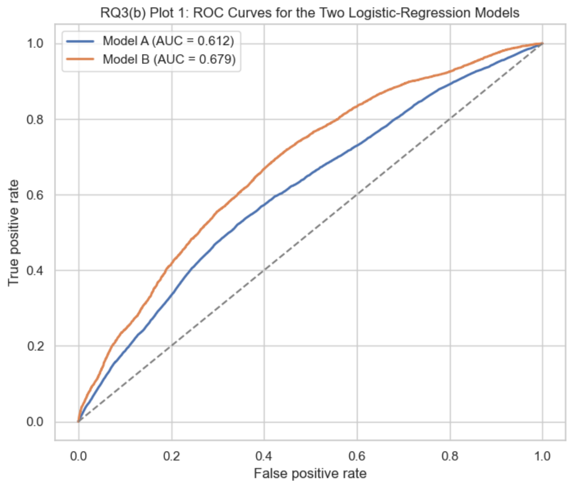

Figure 11: ROC curves comparing the athlete-level logistic regression model against the extended model that includes country/team membership.

The ROC curve for Model B stays above the curve for Model A, and its AUC is higher. This shows that once country/team is added to age, sex, and sport, the model does a better job distinguishing medal-winning from non-medal athlete-event rows.

The improvement is noticeable but not dramatic: both models perform better than random guessing, while Model B provides the stronger overall discrimination. This suggests that country/team membership contains meaningful information about medal probability beyond individual athlete characteristics alone, which directly supports the country-adjusted analysis later.

Limitations:

- ROC curves summarize overall classification performance, but they do not show which specific countries drive the improvement.
- The AUC values are still moderate, so even the better model has limited predictive power.
- Better discrimination does not imply causation; it only shows improved separation between medal and non-medal outcomes.

Adjusted Country/Team Effects from Logistic Regression

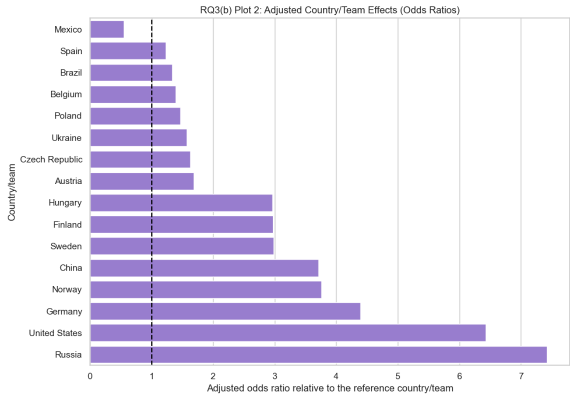

Figure 12: Selected country/team effects from the logistic regression model after controlling for age, sex, and sport category.

This plot shows the adjusted country/team effects from the logistic model. Countries with odds ratios above 1 have higher medal odds than the reference country/team even after controlling for age, sex, and sport category, while countries below 1 have lower adjusted medal odds.

The main result is that country differences still remain after athlete-level characteristics are taken into account. Some countries, such as Russia, the United States, and Germany, show substantially higher adjusted odds, while others fall closer to or below the reference line. This suggests that medal success depends not only on individual athlete characteristics, but also on country-level advantage that remains visible after adjustment.

Overall, the figure provides strong evidence that country/team membership adds important information about medal probability.

Limitations:

- The odds ratios are relative to a chosen reference country/team, so the values are comparative rather than absolute.
- These are model-based associations, not causal effects.
- The plot includes only a restricted set of more frequent countries/teams, so it does not represent every country in the dataset.

Predicted Medal Probability from the Logistic Regression Model

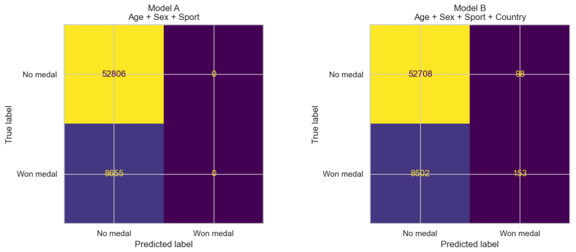

Figure 13: Predicted medal probability from the logistic regression model across athlete characteristics. The plot shows how the model-estimated probability of winning a medal changes across the selected predictor values while holding the other variables fixed at reference levels.

This prediction plot makes the logistic regression results easier to interpret in practical terms because it converts the model output into estimated probabilities rather than only showing coefficients or odds ratios. While the ROC curve shows that the model can distinguish medal-winning from non-medal athlete-event rows overall, this figure shows *how* the predicted probability changes across the athlete characteristics included in the model.

The main pattern is that medal probability does not remain constant across the predictor space. Instead, the estimated probability changes systematically with athlete characteristics such as age, sex, and sport category, which supports the idea that medal success is shaped by more than one factor at a time. This is important for RQ3 because it links the classification model back to the substantive research question: not just whether the model performs better, but what the model suggests about the kinds of athletes or athlete profiles that are more likely to win medals.

This figure also complements the country/team results that follow. Once the athlete-level prediction pattern is established, the country/team effect plot can then be interpreted as showing the remaining differences in medal probability after those baseline athlete characteristics have already been taken into account.

Limitations:

- The predicted probabilities are model-based estimates, so they depend on the logistic regression specification and the chosen reference values for the other predictors.
- If the plot varies one predictor while holding others fixed, it simplifies reality and may hide interactions between variables such as age and sport category.
- The figure shows association, not causation; higher predicted probability does not mean that a predictor directly causes medal success.
- Because the dataset is based on athlete-event rows, repeated participation and team-event structure may still influence the estimated probabilities.

4.12 RQ3 Conclusion and Next Steps

Conclusion (RQ3): The results show that medal success is influenced by both athlete-level and country-level factors. Age, sex, and sport category all have clear associations with the probability of winning a medal, reflecting differences in peak performance ages and competitive structures across sports. The logistic regression results confirm that medal outcomes are appropriately modeled as a binary classification problem. Importantly, extending the model to include country/team membership improves classification performance, as seen in the higher ROC curve and AUC. This indicates that country-level factors—such as training systems, funding, and historical strength—continue to play a meaningful role in medal success even after accounting for individual athlete characteristics.

Next steps (RQ3): We would extend the analysis in several ways. First, we would include interaction terms (e.g., age $\times$ sport category) to capture more detailed relationships between athlete characteristics and medal probability. Second, we would run separate models for major sport groups to account for differences in competitive structure and peak performance ages. Finally, we would incorporate more country-level variables, such as host-country effects and sport specialization, to better explain why countries with similar athlete profiles can have different medal outcomes.

4.13 Conclusion to EDA

Overall, the exploratory analysis shows that Olympic outcomes are shaped by long-term changes in representation, athlete demographics, and country-level participation patterns. Gender parity improves substantially across decades, medalist composition varies by age group and sex, and median medal-winning ages shift over time. At the country level, larger delegations generally win more event medals, but efficiency differs noticeably across countries, showing that participation size alone does not fully explain Olympic success. These exploratory results motivate the use of regression and logistic-regression models in the next stage of the analysis.

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5 Data Analysis

5.1 Analysis Plan

We first cleaned and standardized the Olympics dataset and engineered helper variables such as decade, age groups, binary medal indicators, cleaned country/team names, cleaned sport categories, and an event-medal view for country-level analysis. For RQ1, we used descriptive summaries and visualizations—especially 100% stacked bar charts, decade trend plots, and median/IQR age plots—to compare medal outcomes by age group and sex and to track gender parity and peak-age patterns over time. For RQ2, we aggregated the data to the country-year level and used scatterplots, least-squares regression, log transformation, residual plots, and then multiple regression with `log_participation_size`, `num_sports`, and `year` to model event-medal counts. For RQ3, we followed the course logistic-regression framework and treated medal success as a binary classification problem, modeling `won_medal` using athlete-level predictors such as age, sex, and sport category, and then extending the model with country/team membership. These methods are appropriate because they combine interpretable EDA with regression and classification models that directly match the three research questions.

5.2 Baseline

For **RQ2**, we initially considered a simple least-squares regression using country-year participation size alone to predict event-medal counts. However, this model did not provide an adequate fit, since participation size by itself could not explain enough of the variation in medal outcomes. As a result, we used a multiple regression model with additional predictors, including participation size, number of sports, and year, because it captured the relationship more effectively and produced a better fit to the data.

For **RQ3**, the baseline classifier is an athlete-level logistic regression model using age, sex, and sport category. We then compare this against an extended logistic regression model that also includes country/team membership to test whether country-level advantage improves prediction..

5.3 Main Results

The analysis supports all three research questions.

For **RQ1**, the decade-level parity plots show a substantial increase in female participation over time, moving much closer to parity in recent decades. Medalist composition also varies by age group and sex, and the median/IQR age plots show that typical medal-winning ages shift across decades.

For **RQ2**, country-year participation size is strongly and positively associated with event-medal counts, but the simple least-squares model using participation size alone was not an adequate fit. The raw scatterplots still show the overall upward relationship, but the raw-scale pattern is highly skewed and leaves substantial variation unexplained. Log transformation improves visibility and linearity, and the multiple regression model using `log_participation_size`, `num_sports`, and `year` fits the data better than the one-predictor model. The actual-versus-predicted and residual plots show that the multiple regression explains more of the medal pattern, although some structure and unexplained variation remain. Efficiency and dominance plots also show that countries with similar delegation sizes can differ meaningfully in their ability to convert participation into medals.

For **RQ3**, logistic regression is the appropriate course-aligned method because medal success is binary. The athlete-level model shows that age, sex, and sport category are associated with medal probability, and the extended model with country/team membership improves classification performance further. This indicates that athlete characteristics matter, but country-level advantage still contributes to medal probability after those characteristics are taken into account.

Model / Method	Main metric	Main takeaway
Simple regression	RMSE / R^2	Positive but limited first summary
Multiple regression	Improved R^2	Better fit than simple regression
Logistic regression	AUC / Accuracy	Athlete traits and country improve prediction

If you used a train/test split: We used a train/test split for the logistic-regression classification stage in RQ3 so that model performance was assessed on held-out data rather than training data only.

5.4 Interpretation

Taken together, the results show that Olympic success is shaped by both long-term representation trends and competitive structure at the athlete and country levels.

For **RQ1**, the main interpretation is that Olympic participation and medal patterns have changed substantially over time rather than remaining fixed. The decade-level parity plots show that female participation rose steadily across the historical period and moved much closer to parity in recent decades, which suggests that the Olympic Games became more inclusive over time. At the same time, the age-group medal plots and decade-level age summaries show that medal outcomes are not distributed uniformly across age and sex. Instead, success is concentrated in particular age ranges, and those ranges shift somewhat across decades. In practical terms, this means that Olympic medal patterns reflect both changes in access and changes in the athlete population itself. The results therefore suggest that age structure and sex composition are not just background features of the data; they are part of how Olympic performance is organized and how representation has evolved historically.

For **RQ2**, the results show that participation size matters, but it is not the whole story. Countries that send more athlete-event entries generally win more event medals, so larger delegations do have a structural advantage in absolute medal totals. However, the raw scatterplots also show that countries with similar participation sizes can end up with noticeably different medal outcomes, which means delegation size alone does not fully determine success. The log-transformed plots and multiple regression results reinforce this point: the overall relationship is clearly positive, but a better-fitting model is needed because participation size by itself leaves too much unexplained variation. The efficiency rankings deepen that interpretation by showing that some countries convert participation into medals much more effectively than others. The dominance plots then show that repeated Olympic leadership is concentrated among a relatively small set of countries. Overall, the interpretation for RQ2 is that Olympic dominance reflects both *scale* and *conversion efficiency*. Large delegations are more likely to produce high medal totals, but efficient countries can outperform others with similar levels of participation, so raw medal counts should not be treated as a complete measure of national performance.

For **RQ3**, the results show that medal probability depends on a combination of athlete-level and country-level factors. The grouped sport-category plots indicate that medal rates differ across sport types and also vary somewhat by sex, which suggests that the competitive structure of the sport matters for understanding success. The logistic regression results then formalize that pattern: age, sex, and sport category all help explain whether an athlete-event row results in a medal. The extended model with country/team membership performs better than the athlete-only model, which means that country/team contains additional predictive information even after athlete characteristics have already been accounted for. The adjusted country/team odds-ratio plot makes that result more concrete by showing that some countries retain substantially higher or lower medal odds relative to the reference country/team after adjustment. The predicted-probability plots further show that medal chances vary systematically across athlete profiles rather than remaining constant. In plain terms, this means that success is not driven only by who the athlete is; it is also shaped by the broader system they compete within, including country-level advantages that remain visible after adjustment.

Across all three research questions, the overall conclusion is that Olympic outcomes cannot be understood using a simple perspective. **RQ1** shows that representation and age structure change historically. **RQ2** shows that country success depends partly on participation scale, but also on efficiency and concentration of strength. **RQ3** shows that medal success is best understood as a probability shaped by multiple athlete characteristics and by country. Together, these findings suggest that Olympic success is produced by the interaction of demographic change, national participation patterns, and competitive advantage rather than by any one variable alone.

The main limitations of these interpretations should also be kept in mind. The dataset is based on athlete-event rows rather than unique athletes, so participation-based measures can overstate scale when the same athlete appears multiple times. Team events can inflate medal-related counts unless the event-medal view is used. In addition, the grouped sport categories improve interpretability but can hide meaningful within-sport differences. Finally, the regression and logistic-regression results are associative rather than causal, so they identify patterns and predictive relationships, not definitive causal mechanisms.

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6 Real-World Application

Stakeholders. The main stakeholders are national Olympic committees, sport federations, coaches, athlete development programs, and sports analysts.

What decision could they make based on your results? They could use these findings to make better decisions about talent development, event specialization, and how to evaluate country success more fairly than by medal totals alone.

Recommendations.

- **Use age- and sex-specific development pipelines:** Since medal outcomes and peak ages vary by age group and sex, athlete development should be tailored rather than based on a single generic model of peak performance.
- **Do not rely only on medal totals:** Because participation size strongly affects event-medal counts, countries should also be evaluated using medal efficiency and broader participation context.
- **Combine athlete-level and country-level planning:** Logistic regression results suggest that country/team membership still adds predictive value after accounting for athlete characteristics, so structural support systems matter alongside individual talent.

Confidence. We are moderately confident in these recommendations because the major relationships are consistent across multiple plots and models, although the results remain observational and depend on the structure of the dataset.

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7 Report Quality and Reproducibility

7.1 How to run the code

- The project code is available on GitHub: https://github.com/PrinceKanani16/Data2010-olympics-project
- Main file/notebook: `notebooks/Data2010_Project.ipynb`
- Steps to run:
  - Clone or download the GitHub repository.
  - Download the Olympics CSV dataset from Kaggle.
  - Open `notebooks/Data2010_Project.ipynb`.
  - Update the file path in the data-loading cell so it points to your local copy of `olympics_1896_2024.csv`, or move the file into the repository `data/` folder and use that path instead.
  - Run all notebook cells in order from top to bottom.
- Required packages: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn`, `statsmodels`

7.2 Project Structure

- `README.md` (project overview and instructions for running the code)
- `notebooks/` (main analysis notebook)
- `data/` (dataset used in the project)
- `figures/` (exported plots organized by research question)
- `outputs/` (tables and model outputs)
- `report/` (final written report)

The project was organized into separate folders for the dataset, analysis notebook, exported figures, generated tables, and final report to improve readability and reproducibility. The main analysis was completed in a Jupyter notebook, while figures were saved into subfolders by research question. Tables model outputs were stored separately so they could be referenced more easily in the report and appendix.

```
Project/figures/
|
|-- notebooks/
|   \-- Data2010_Project.ipynb
|
|-- data/
|   \-- olympics_1896_2024.csv
|
|-- figures/
|   |-- RQ1_plots/
|   |-- RQ2_plots/
|   |-- RQ3_plots/
|   \-- Appendix/
```

```
|
|-- tables/
|
|-- report/
|   \-- DATA2010_Project_Report.tex
|
|-- other
```

---

8 Contributions (If Group Project)

- Prince Kanani: selection and refinement of the research questions; completed most of the data cleaning and feature engineering; developed the sport-category grouping; built and organized the main notebook analysis; created and coded the visualizations; selected figures for the final report; carried out the regression and logistic-regression modeling and model comparison; wrote major parts of the notebook interpretations, conclusions, and limitations; and contributed to reproducibility through GitHub organization and template integration.
- Joshua Mudiwa: defining the project overview and scope; supported data cleaning and country/team grouping; helped organize figures and appendix exploration; contributed to the data source and description section; supported the data-analysis write-up; worked on report formatting, structure, and presentation; and contributed to conclusions, limitations, next steps, and overall report quality.

The project was completed collaboratively. Although each member had primary responsibility for the tasks listed above, all major parts of the work were discussed and cross-checked together through in-person meetings.

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9 Conclusion and Next Steps

Conclusion: This project shows that Olympic performance and representation changed substantially from 1896 to 2024. Female participation moved much closer to parity over time, medalist composition varied across age and sex groups, and typical medal-winning ages shifted across decades. At the country level, larger delegations generally won more event medals, but efficiency and dominance varied considerably, showing that participation size alone does not fully explain medal success. Finally, athlete-level logistic regression showed that medal probability depends on age, sex, and sport, and that country/team membership still improves prediction, indicating that country-level advantage remains important after athlete characteristics are taken into account.

Next steps: If we had one more week, we would improve the project in five ways:

- Separate Summer and Winter Games more systematically
- Extend the peak-age and medal-probability analyses within individual sport groups,
- Report a fuller set of logistic-regression performance metrics and threshold choices,
- Test additional country-level predictors such as host-country effects or more refined sport-specialization variables.
- Compare team sports and individual sports more explicitly, since medal structures differ a lot between them

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10 AI Disclosure

- We primarily used ChatGPT for generating prompts, checking code, and supporting analysis, with limited use of Claude for additional code review and analysis.
- We used them for:
  - Olympic dataset cleaning, organization, and presentation
  - Choosing the most appropriate plots for each research question to improve visualization
  - Using prompts to strengthen the final report's writing, structure, and formatting
  - Debugging code when we encountered errors
- Some examples of the prompts we used were the following:
  - Which of these research questions would be better for our project?
  - Where is the best place in this project to use a violin plot?
  - Improve this sentence for the report.

– Why are there spaces throughout the document

References

- Olympic Summer and Winter Games (1896–2024). Kaggle dataset. Available at: https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024. Accessed March 25, 2026.
- Course notes: We used the lecture notes on Pandas, Feature Engineering, Visualization 1 and 2, EDA, Simple Linear Regression, Logistic Regression, Multiple Regression and Cross Validation University of Manitoba, Winter 2026. https://pandas.pydata.org/
- scikit-learn documentation. Available at: https://scikit-learn.org/stable/
- statsmodels documentation. Available at: https://www.statsmodels.org/

A Appendix

Extra figures, extra tables, or small code snippets that would clutter the main report.

A.1 Top 15 Countries after Cleaning the Dataset

Table 1: Top countries by participant count after dataset cleaning

Country	Count
United States	16681
Germany	14396
France	10756
United Kingdom	10691
Russia	10042
Italy	9722
Canada	9057
Japan	8212
Australia	7566
Sweden	7368
Hungary	6042
Poland	5890
Czech Republic	5549
Switzerland	5534
Netherlands	5440

A.2 Total Number of Participants in Each Olympic Sport Category

Table 2: This table shows the total number of participants in each Olympic sport category after cleaning the dataset and regrouping individual sports into broader categories (Feature Engineering). It highlights how athlete participation is distributed across categories, with Winter Sports and Athletics / Urban having the largest counts.

Sport Category	Count
Winter Sports	40548
Athletics / Urban	39563
Aquatics	29142
Combat	27964
Gymnastics	22240
Team Sports	21994
Boating / Rowing	20248
Precision Sports	13632
Strength / Mixed	11905
Cycling	10338
Racquet Sports	6725
Other / Historical	1545

A.3 Proportion of Olympic Medals Won by Continent

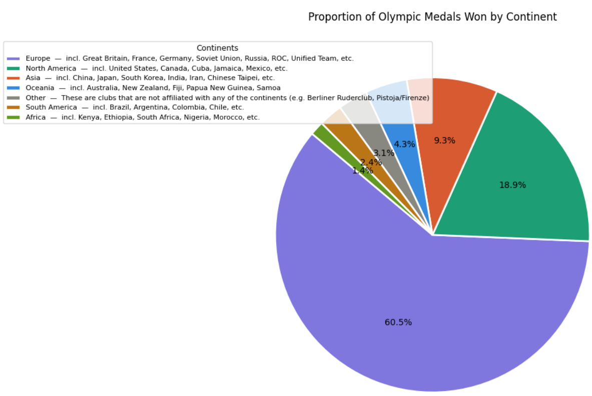

Figure 14: Supplementary figure showing the distribution of medal-winning entries across continents.

A.4 Medal Type Share by Age Group (Gold/Silver/Bronze)

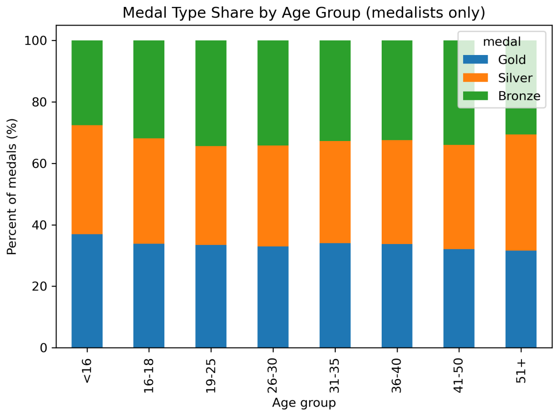

Figure 15: Each bar sums to 100%, showing how medals within each age group split into Gold, Silver, and Bronze.

A.5 Share of Olympic Years Dominated by Country (2000–2024)

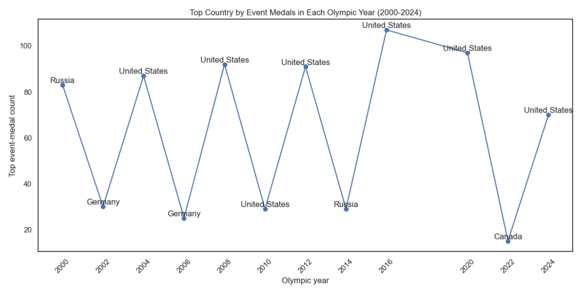

Figure 16: Country comparison for a selected Olympic year, showing how dominance relates to delegation size and medal efficiency.

A.6 RQ2 Multiple Regression Coefficients

Table 3: Coefficient estimates from the multiple regression model used in Research Question 2. Positive coefficients indicate predictors associated with higher log-transformed event-medal counts, holding the other variables constant.

Predictor	Coefficient	p-value
Intercept	8.518	< 0.001
$\log(1 + participationsize)$	0.340	< 0.001
Number of sports	0.068	< 0.001
Year	-0.0047	< 0.001

A.7 RQ3 Logistic Regression Model Comparison

Table 4: Prediction-performance comparison for the two logistic-regression models in Research Question 3. Model A uses athlete-level predictors only, while Model B adds country/team membership.

Metric	Model A	Model B
Specification Or Parameters	Age + sex + country/team	Model A + sport category
Threshold	0.50	0.50
Accuracy	0.859	0.860
Precision	0.000	0.610
Recall	0.000	0.018
F1 score	0.000	0.034
ROC AUC	0.612	0.713

A.8 Top 25 Countries by Medal Efficiency

Table 5: Top 25 countries ranked by event-medal efficiency, measured as event medals per 100 participation rows after applying the minimum-size filter.

Rank	Country	Participation	Event Medals	Medals/100
1	Russia	10,042	1,873	18.65
2	United States	16,681	2,939	17.62
3	Azerbaijan	383	59	15.40
4	Ethiopia	408	61	14.95
5	Kenya	801	116	14.48
6	China	5,171	740	14.31
7	Georgia	356	49	13.76
8	Germany	14,396	1,819	12.64
9	Norway	4,492	482	10.73
10	Cuba	2,221	237	10.67
11	Jamaica	929	92	9.90
12	Iran	753	74	9.83
13	Finland	4,967	485	9.76
14	Armenia	259	24	9.27
15	Sweden	7,368	669	9.08
16	Hungary	6,042	546	9.04
17	UK	10,691	944	8.83
18	South Korea	3,912	336	8.59
19	Indonesia	444	38	8.56
20	France	10,756	901	8.38
21	Netherlands	5,440	448	8.24
22	Italy	9,722	789	8.12
23	Uzbekistan	643	52	8.09
24	North Korea	693	56	8.08
25	Romania	3,880	312	8.04

A.9 Top Event-Medal Country by Olympic Year

Table 6: Top event-medal country in each Olympic year included in the dominance analysis.

Year	Country	Event Medals
2000	Russia	83
2002	Germany	30
2004	United States	87
2006	Germany	25
2008	United States	92
2010	United States	29
2012	United States	91
2014	Russia	29
2016	United States	107
2020	United States	97
2022	Canada	15
2024	United States	70